# Data mining project

This project analyzes crimes records from San Francisco to identify patterns and propose data-driven strategies for crime reduction. The dataset includes:

- Crime types (e.g., theft, assault, vandalism).
- Coordinates (X, Y).
- Different types of time (time, day, holidays).

We chose to work on the **'Crime rate data '**because we have already worked on a dataset similar to the customers preferences, so we liked to explore a new type of data to work on.

And with the continuous rise in crime rates in many cities around the world, the need to analyze and understand the nature of crime has become more critical than ever to help reduce crime rates.

## Data preprocess:

- Handling Holidays/Weekends: Marked holidays (is_holiday) and weekends (is_weekend) to analyze seasonal trends.
- Feature Engineering: Dropped rare crime categories (<1000 crime) and merged low-frequency crimes into "others".

## Data analysis:

- We split the data into clusters using k-medoids to use the data to predict any possible future crimes.
- We visualized some data like: the number of crimes per year, most frequent crime per year ….etc.
- We did multiple visualizations to display different perspectives of the data like: most frequent crime per year, and most frequent crime per year (without repeating the same crime twice in the graph).
- We tried to use a new type of visualization to display the locations of the crimes on a map.

**Key points found:**

- Hotspots: Crimes concentrated in zones with X > 0.5 (for example: downtown areas).
- Top 5 Crimes: Larceny/Theft, Other Offenses, Non-Criminal, Assault, Drug/Narcotic.
- **Peak times:**
  - Theft/Vandalism: Highest at Time ≈ 21.5 (9:30 PM) and weekends.
  - Assaults: Frequent on Saturdays.

We found some rare crimes, so we dropped them to avoid any outliers or skewing results. And we used StandardScaler to normalize Time, X, and Y for clustering.

This is just a sample of what we did on our project, and we will display all the remaining codes, visualizations, and analysis on our Jupyter notebook.